CoRL RoboMem Workshop 2026

GEMA: Generative Memory for Robotic Manipulation

First Author · Second Author | Institution Name

PAST EXPERIENCE

Remember what happened.

TASK-SPECIFIC GUIDANCE

Generate what should happen next.

01 Motivation, insight & pipeline

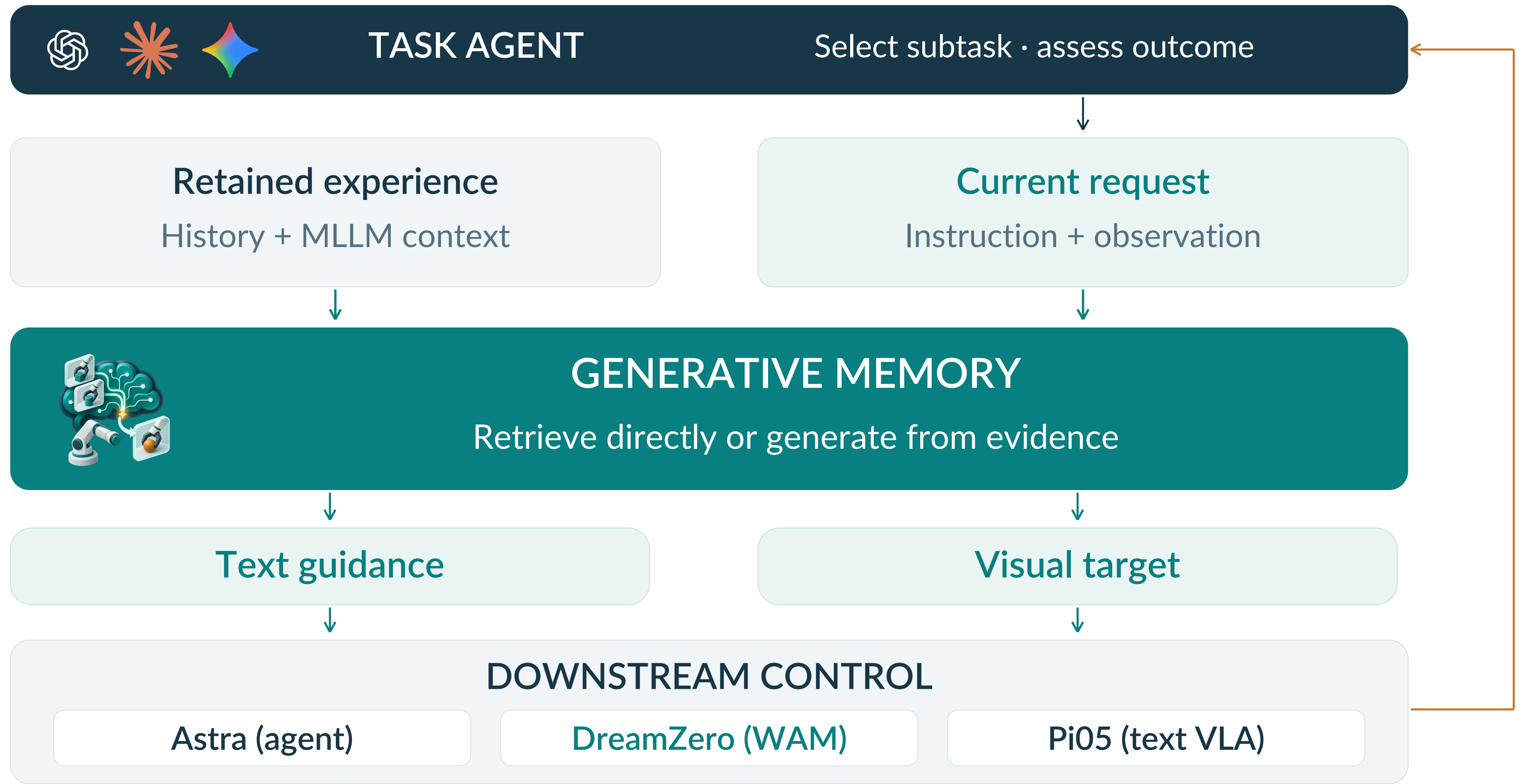
MOTIVATION

Memory needs change as a long-horizon task unfolds.

Earlier locations, relations or instructions may no longer be visible.

INSIGHT · Inspired by constructive episodic memory

An MLLM turns retained experience into task-specific references for control.



TASK AGENT

Select subtask · assess outcome

Retained experience

History + MLLM context

Current request

Instruction + observation

GENERATIVE MEMORY

Retrieve directly or generate from evidence

Text guidance

Visual target

DOWNSTREAM CONTROL

Astra (agent) | DreamZero (WAM) | Pi05 (text VLA)

Execute → observe → assess → update memory

Query: place the cube 10 cm right of its original position

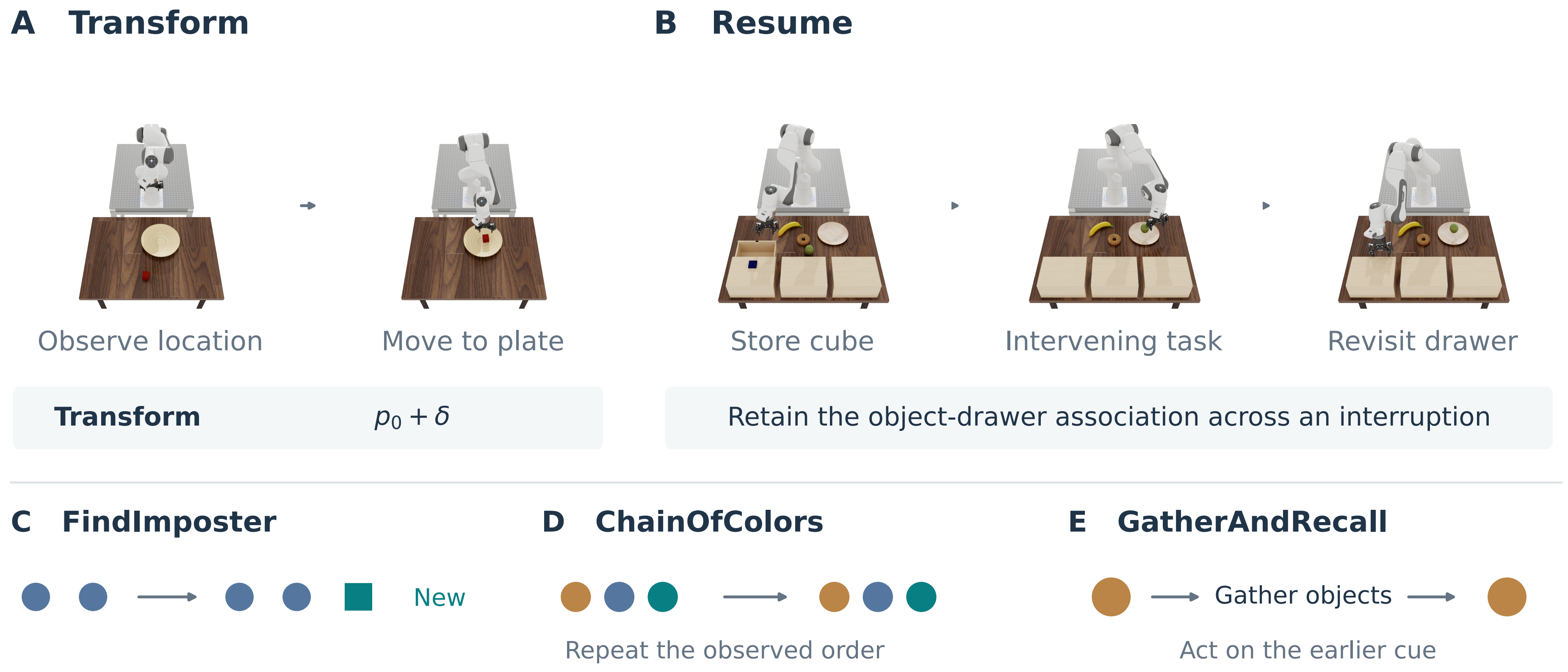
Observed anchor · Same historical anchor · Instruction-dependent target

Retrieve the original position; generate the offset.

02 Benchmark design

A. TASK-BASED

Nine controlled tasks test recall, historical comparison and construction of new targets.



Transform $p_0 + \delta$

Retain the object-drawer association across an interruption

Repeat the observed order

Act on the earlier cue

Transform + Resume + seven MIKASA variants

B. CONTINUOUS-QUERY (Con-Q)

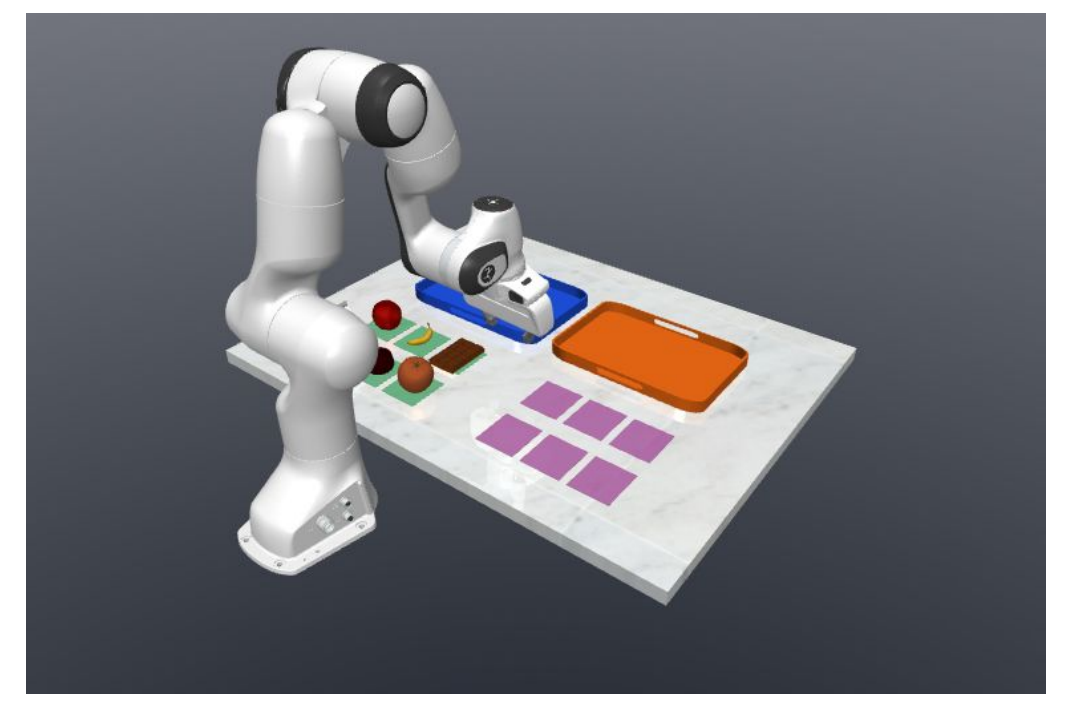
Persistent scenes with procedurally generated queries.

Sequential

Event order / task resumption

Spatial

Past locations / drawer associations



Semantic

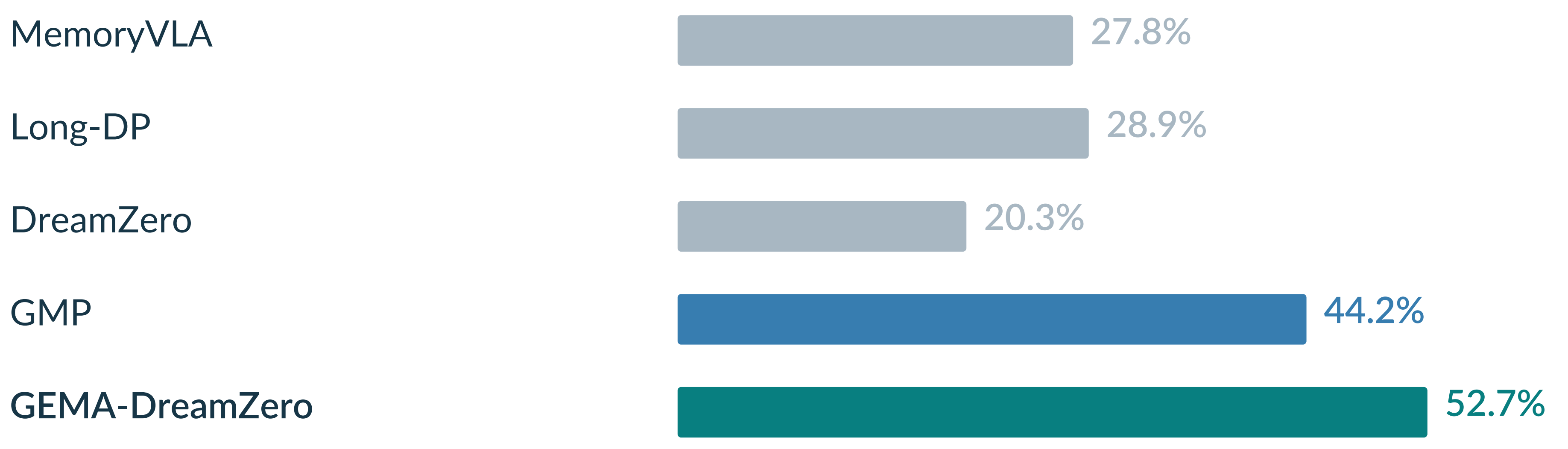
Order contents / past associations

Query → act → update scene → next query · no reset

03 Results

A. TASK-BASED

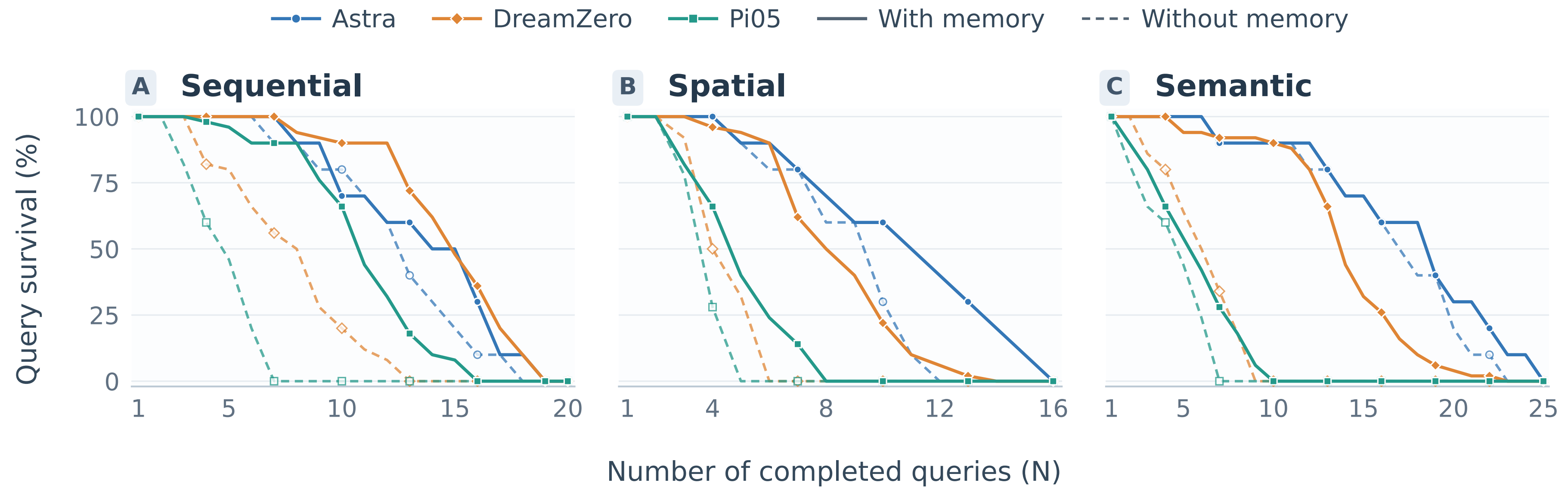
Mean task success (%) across nine tasks



100 evaluation episodes per task; unweighted mean.

B. Con-Q

Sequential / spatial / semantic query survival



$$S(N) = \frac{\text{Rollouts completing at least N queries}}{\text{Total started rollouts}} \times 100\%$$

No query resets; stop at first failure or 15 min of simulation.

Rollouts: Astra 10; DreamZero / Pi05 50 each per setting.